

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_____ show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129 _____

Does R2 show up as an OSPF neighbor on R1? _____ Yes _____

Does R2 show up as an OSPF neighbor on R3? _____ No _____

What does this information tell you?

_____ Traffic from R3 to the 192.168.2.0/24 network will still go through R1. Since R2's S0/0/1 interface is passive, it does not advertise OSPF updates. The total cost of 129 comes from two T1 serial links (64 each) plus the Gigabit 0/0 link on R2 (1). _____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____ R2(config)# router ospf 1 _____

_____ R2(config-router)# no passive-interface s0/0/1 _____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/1 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____ 65 _____ comes from one T1 serial link costing 64 and the R2 Gigabit 0/0 LAN link costing 1. _____

Is R2 listed as an OSPF neighbor to R3? _____ Yes _____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_____ Modern devices can operate at speeds above 100 Mb/s, so a higher default reference bandwidth is required to calculate link costs more accurately. _____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_____ 192.168.3.0/24: R1 S0/0/1 + R3 G0/0, $781+1=782$ _____

_____ 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1, $781+64=845$ _____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_____ 1562 _____

_____ Each serial link has a cost of 781, and the route to 192.168.23.0/24 has two serial links. _____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

_____ OSPF selects the path with the lowest total cost. In this case, the best route is through **R1-S0/0/0, R2-S0/0/1, and R3-G0/0**, with a total cost of **1,563**. This is lower than the alternate path through **R1-S0/0/1 and R3-G0/0**, which has a cost of **1,566**. _____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router IDs affect the DR and BDR election on multi-access networks. If a router ID is tied to a physical interface, it may change when that interface goes down. To avoid this, use a loopback interface IP address or configure it manually with the router-id command.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR election matters only on multi-access networks like Ethernet or Frame Relay. Since the serial links in this lab are point-to-point, no DR/BDR election takes place.

3. Why would you want to set an OSPF interface to passive?

Setting a LAN interface as passive stops unnecessary OSPF updates from being sent on that interface, which saves bandwidth. The router will still advertise the network to neighboring routers.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF areas allow a hierarchical network design that reduces routing overhead and improves scalability.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The OSPF route will be selected because it has a lower administrative distance (110) than RIP (120) and EIGRP (internal 90, external 170), making it the preferred routing option.

EIGRP

0